

Distributed Simulation Framework

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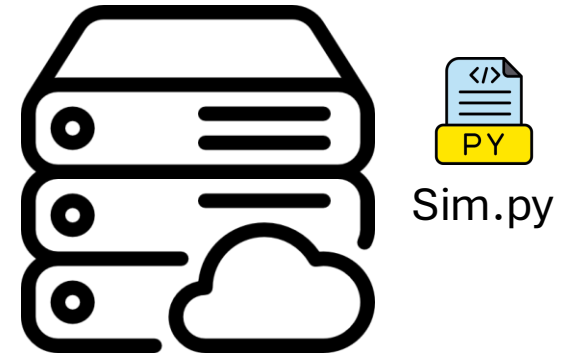
NWSL, Computer Engineering, UCF

7th March, 2025

How do we run simulation ?

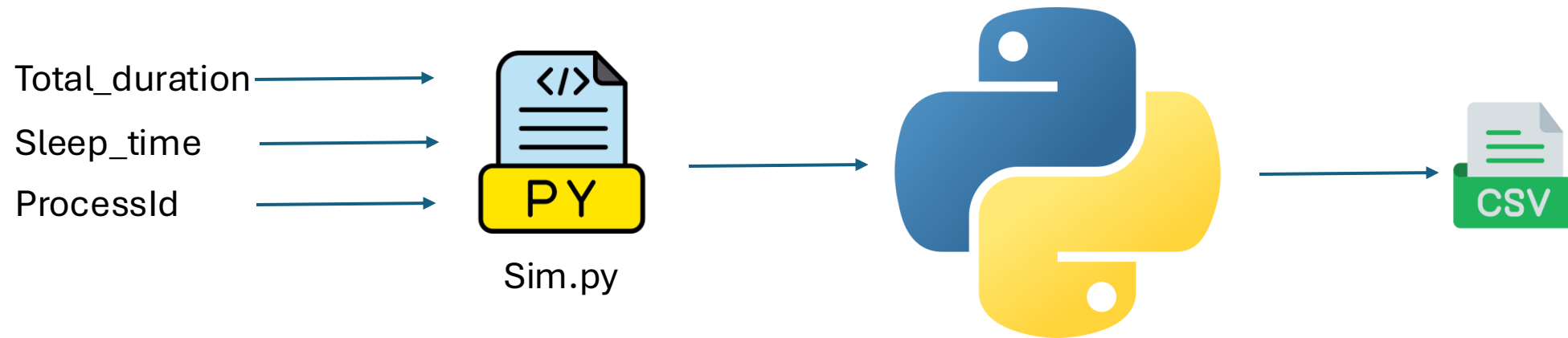


Personal Computer



ARCC

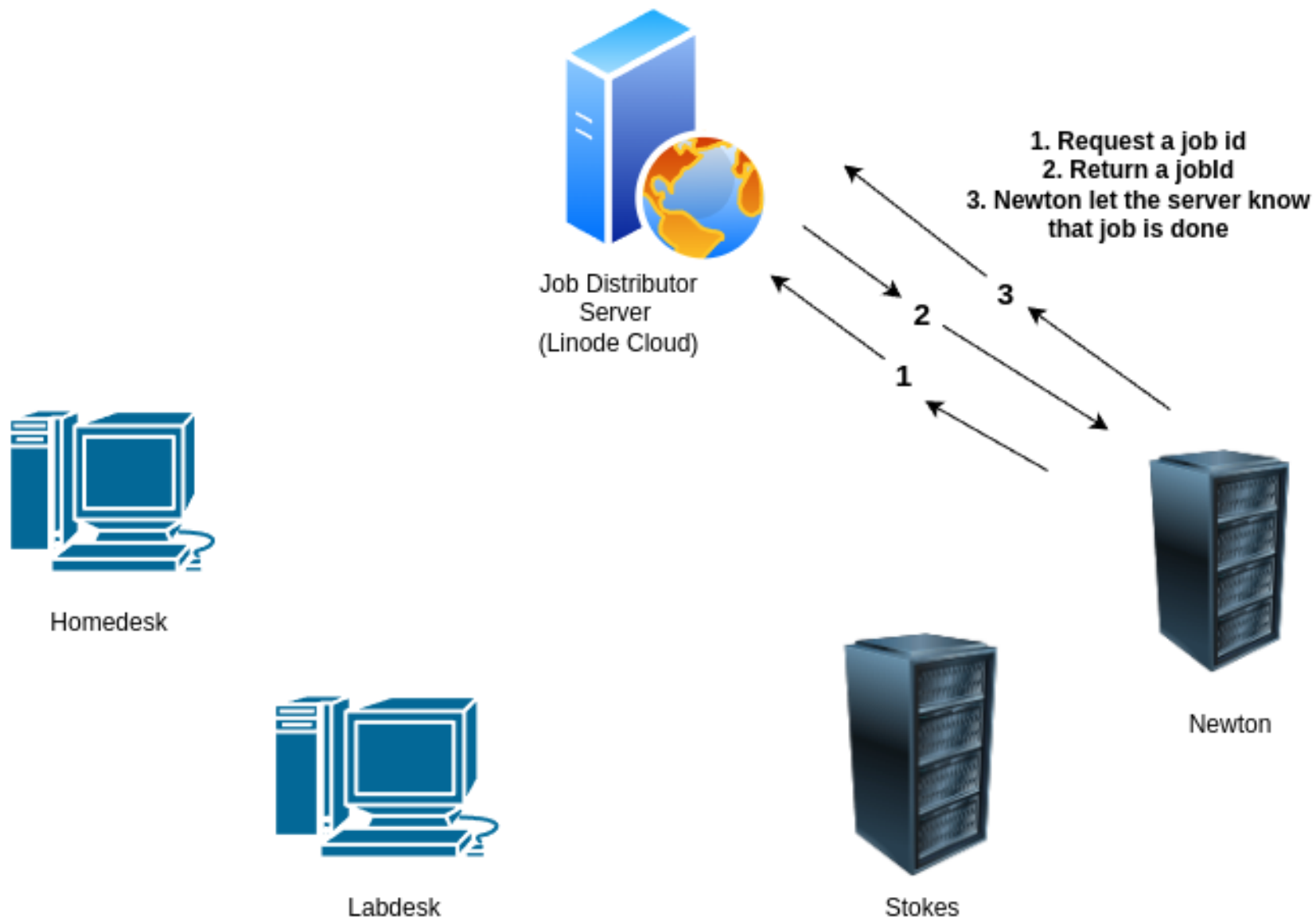
Suppose, you want to run the simulation 5 times for 5 different parameter. How would you do that?



```
python sim.py --total_duration=100 --sleep_time=10 --processId=3
```

Problem ??

- Your PC could crash due to insufficient memory.
- A specific process might fail because of a runtime error.
- You may need to use your PC for other tasks.
- You may want to add more computational resources in your simulation.



Job State

- NOT_STARTED
- SERVED
- ABORTED
- DONE



Homedesk



Labdesk



Job Distributor
Server
(Linode Cloud)

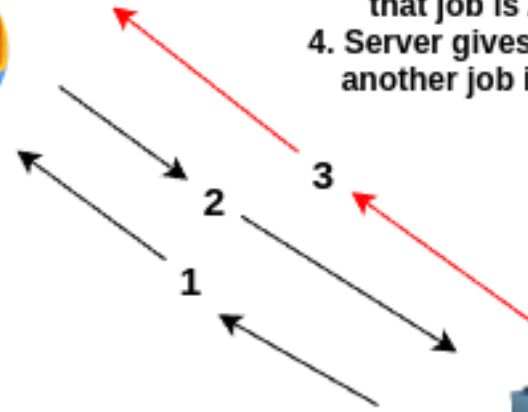


Stokes



Newton

- ABORTED Case**
1. Request a job id
 2. Return a jobid
 3. Newton let the server know that job is Aborted
 4. Server gives the newton another job id (Step 2)





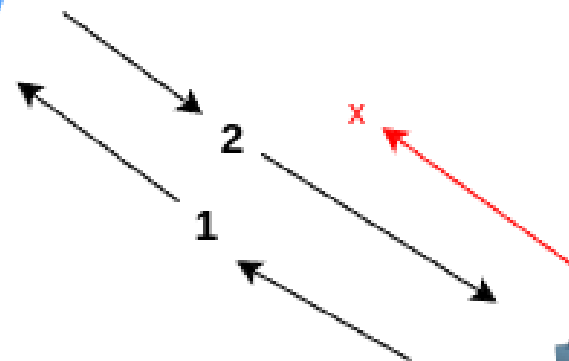
Homedesk



Labdesk



Job Distributor
Server
(Linode Cloud)



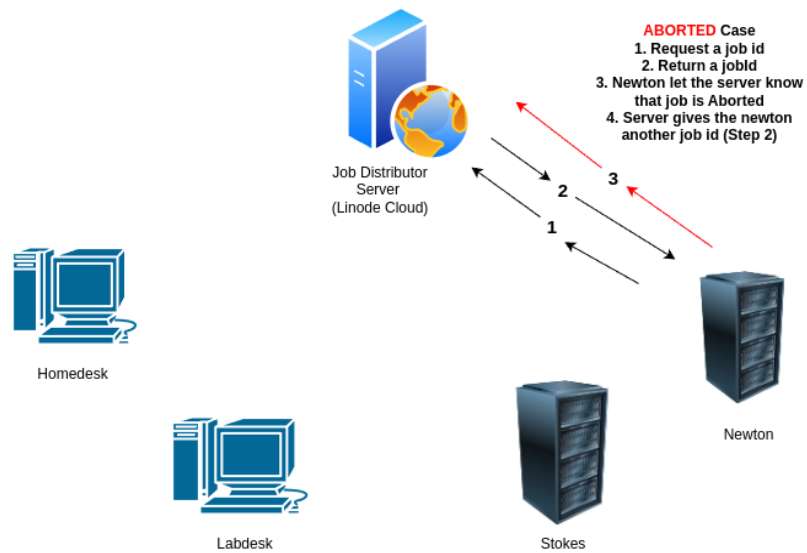
No Response Case
1. Request a job id
2. Return a jobid
3. Newton node has been failed or labdesk goes off.



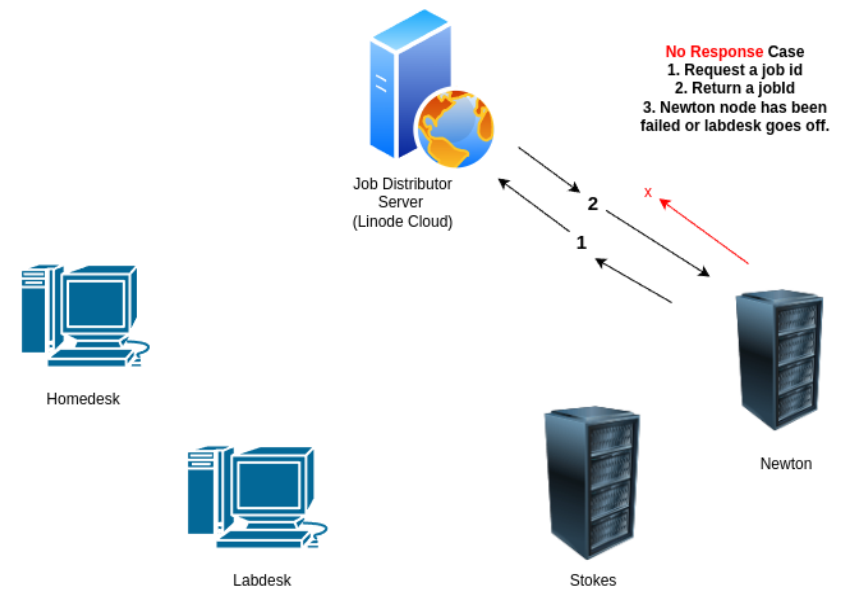
Newton



Stokes



ABORTED jobs



SERVED jobs but no response for a while

